

NuMicro[®] Family VSCode Quick Start Guide

The information described in this document is the exclusive intellectual property of Nuvoton Technology Corporation and shall not be reproduced without permission from Nuvoton.

Nuvoton is providing this document only for reference purposes of NuMicro microcontroller based system design. Nuvoton assumes no responsibility for errors or omissions.

All data and specifications are subject to change without notice.

For additional information or questions, please contact: Nuvoton Technology Corporation.

www.nuvoton.com

TABLE OF CONTENTS

1 INSTALLATION VSCODE AND EXTENSIONS 3

2 GET STARTED WITH AN EXAMPLE PROJECT 4

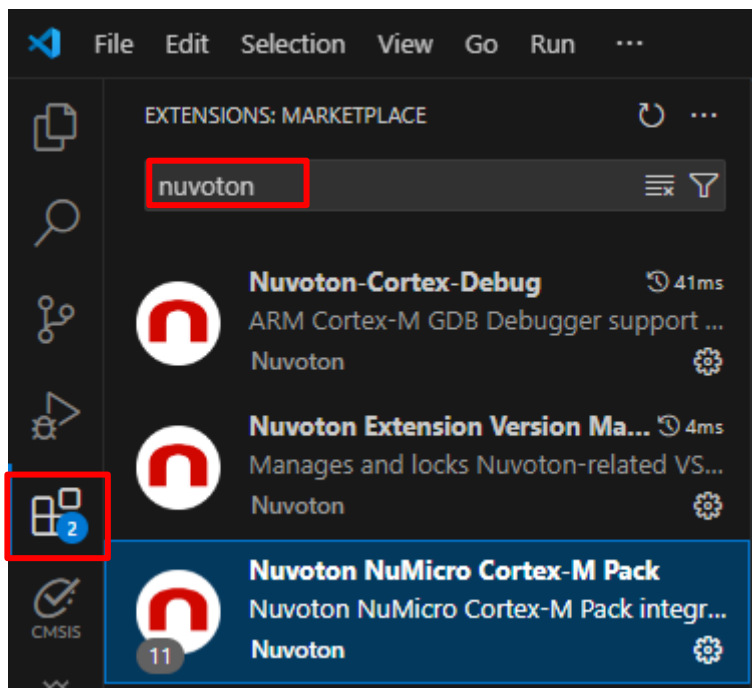
3 MANAGE ARM LICENSE 8

4 CONFIGURE THE DEVICE 10

5 RUN THE EXAMPLE PROJECT 12

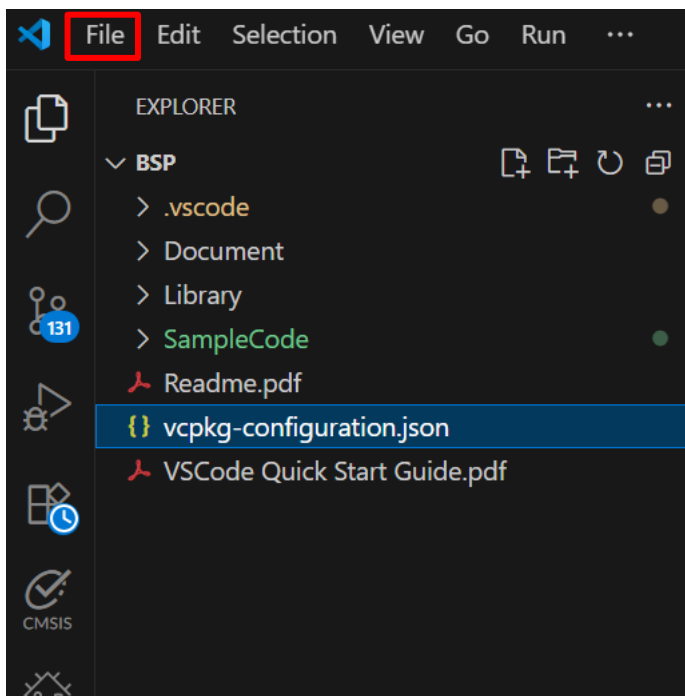
1 INSTALLATION VSCODE AND EXTENSIONS

1. Download VSCode from <https://code.visualstudio.com> and install it.
2. Launch VSCode and click Extensions in the Activity Bar.
3. Text “**Nuvoton NuMicro Cortex-M Pack Extension**” in search bar. Click install it.

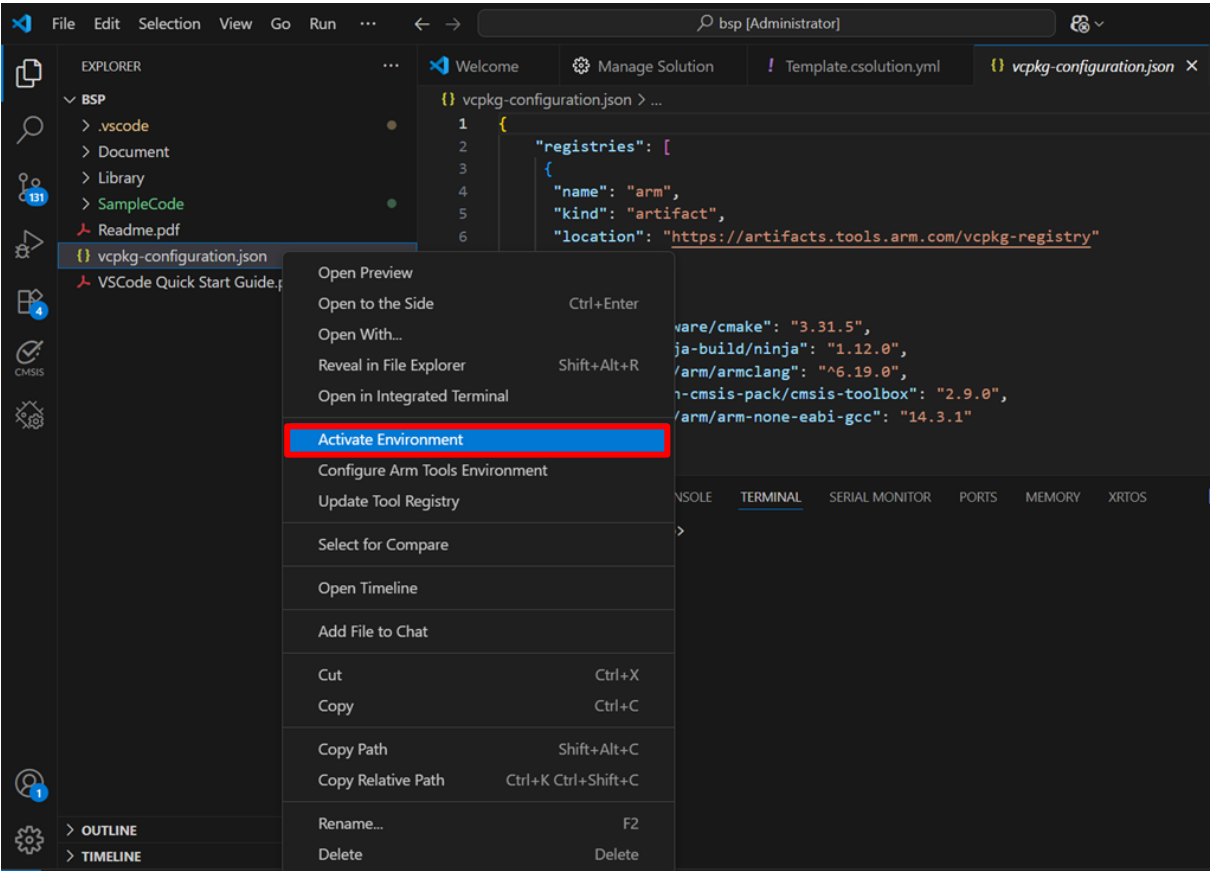


2 GET STARTED WITH AN EXAMPLE PROJECT

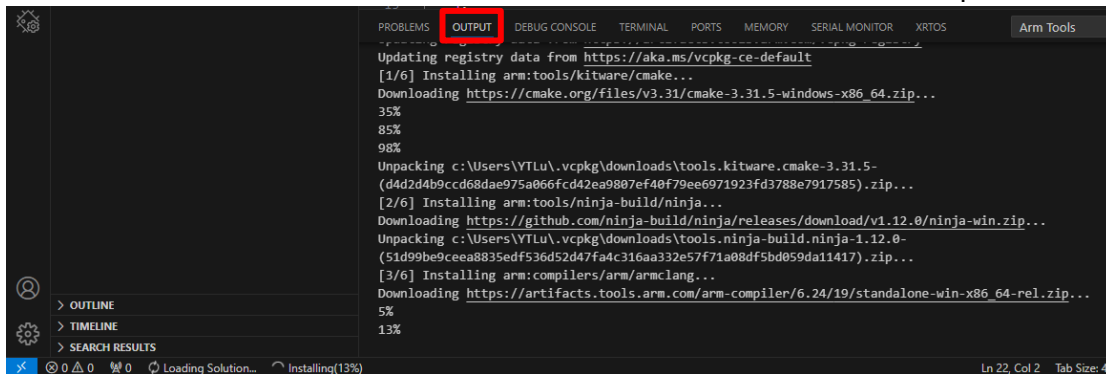
1. Click **File** and select "**Open Folder**" in the toolbar. Then select the path of the example project as below.



2. Right-click `vcpkg-configuration.json` and select **Activate Environment**. (Please use private network connection)



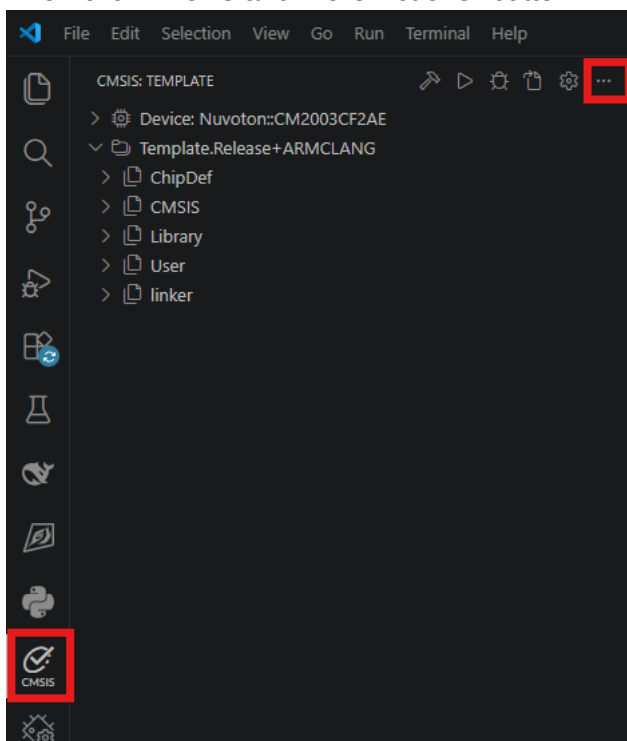
3. Check OUTPUT terminal at the bottom. It will download and install the requires tools.



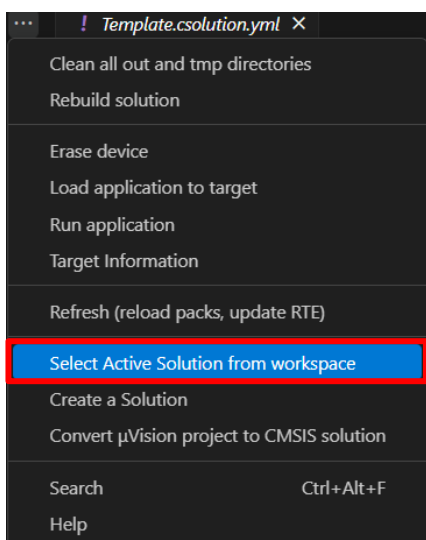
The screenshot shows the OUTPUT terminal in VS Code with the following text:

```
Updating registry data from https://aka.ms/vcpkg-ce-default
[1/6] Installing arm:tools/kitware/cmake...
Downloading https://cmake.org/files/v3.31/cmake-3.31.5-windows-x86_64.zip...
35%
85%
98%
Unpacking c:\Users\YTLu\.vcpkg\downloads\tools.kitware.cmake-3.31.5-
(d4d2d4b9ccd68dae975a066fcd42ea9807ef40f79ee6971923fd3788e7917585).zip...
[2/6] Installing arm:tools/ninja-build/ninja...
Downloading https://github.com/ninja-build/ninja/releases/download/v1.12.0/ninja-win.zip...
Unpacking c:\Users\YTLu\.vcpkg\downloads\tools.ninja-build.ninja-1.12.0-
(51d99be9ceea8835edf536d52d47fa4c316aa332e57f71a08df5bd059da11417).zip...
[3/6] Installing arm:compilers/arm/armclang...
Downloading https://artifacts.tools.arm.com/arm-compiler/6.24/19/standalone-win-x86_64-rel.zip...
5%
13%
```

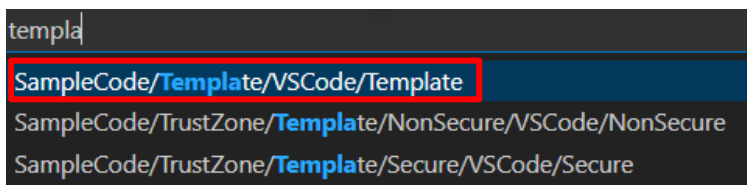
4. Click CMSIS in the Activity Bar. Then click “Views and More Actions” button.



5. Select Active solution from workspace.

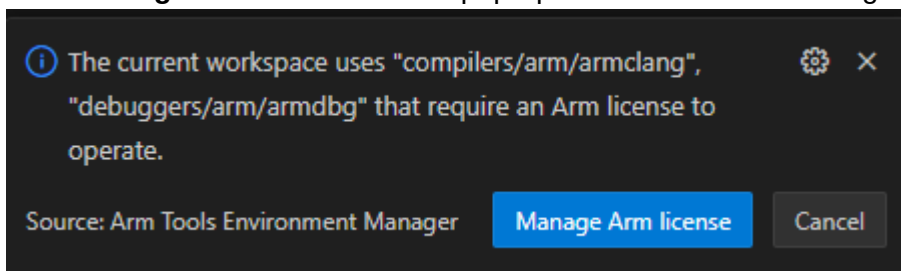


6. Select an example code to activate it.

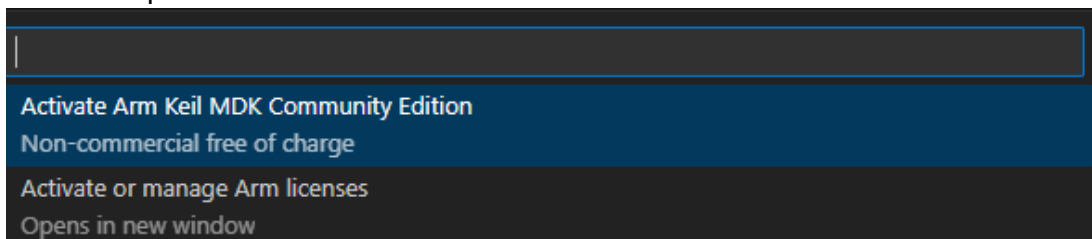


3 MANAGE ARM LICENSE

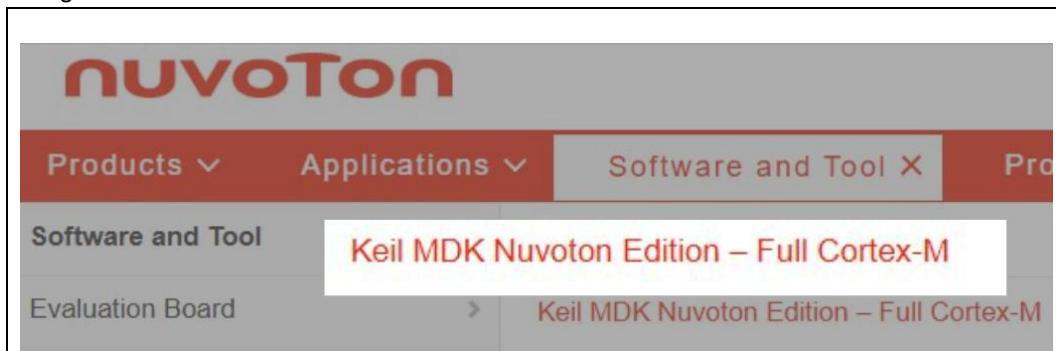
1. Click **Manage Arm License** in the pop-up window at the bottom right.



2. In the search bar at the top, select "**Activate or manage Arm licenses**" from the available options.

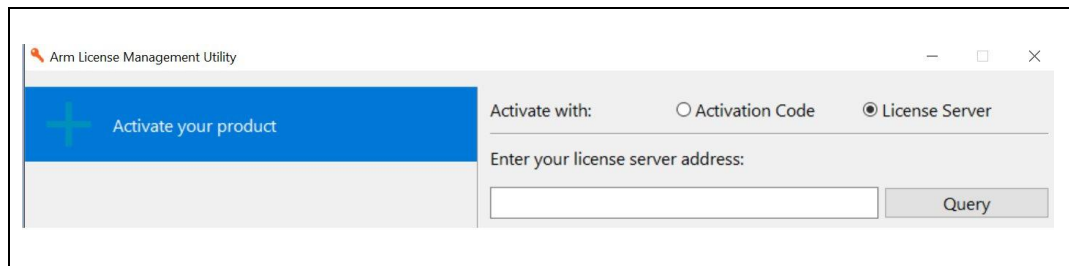


3. Get Keil MDK License ID code
 - a. Navigation to Official Website



- b. Fill Out the Form

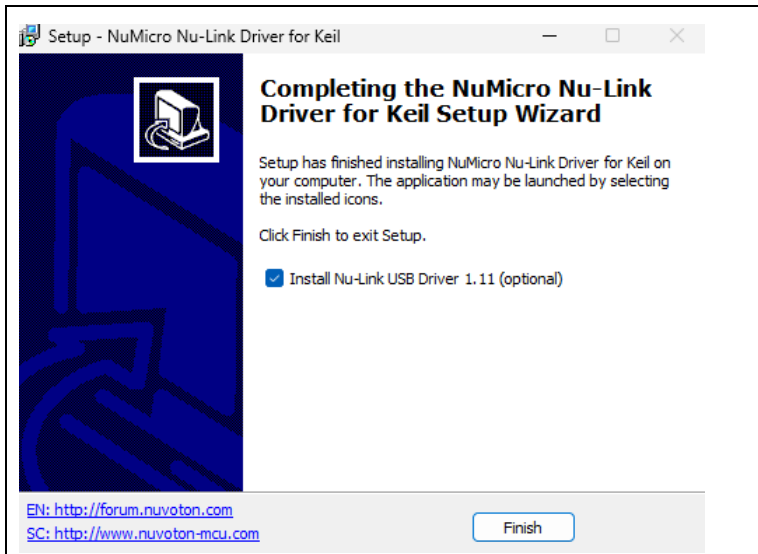
c. Check Mailbox and Fill in the License Server



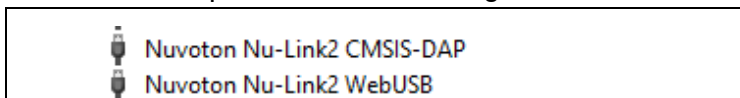
The screenshot shows a window titled "Arm License Management Utility". On the left, there is a blue button with a green plus icon and the text "Activate your product". On the right, there are two radio buttons under the heading "Activate with:". The first is "Activation Code" (unselected) and the second is "License Server" (selected). Below these, there is a text input field labeled "Enter your license server address:" and a "Query" button.

4 CONFIGURE THE DEVICE

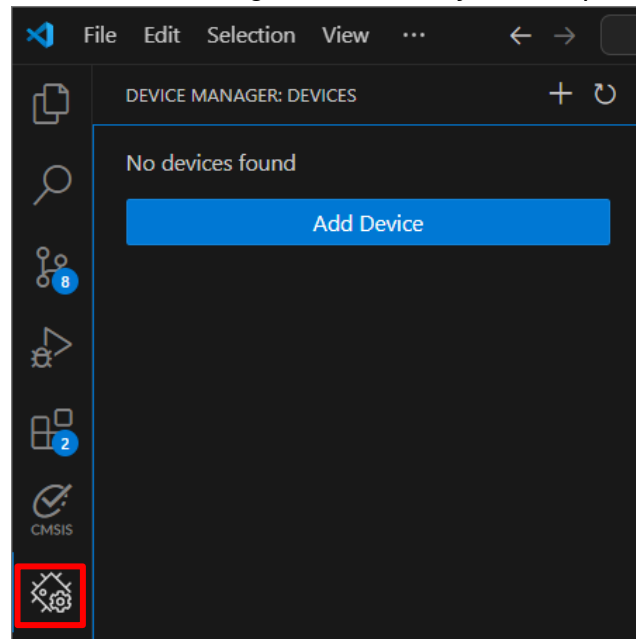
1. Install Nuvoton Nu-Link Keil Driver
2. After installing the Keil driver, please check the box to install the Nu-Link USB Driver.



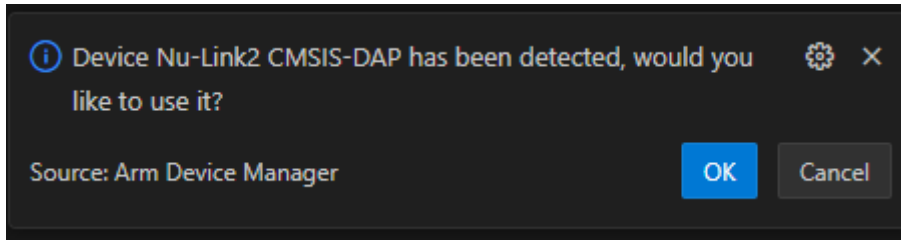
3. Installation complete in Device Manager



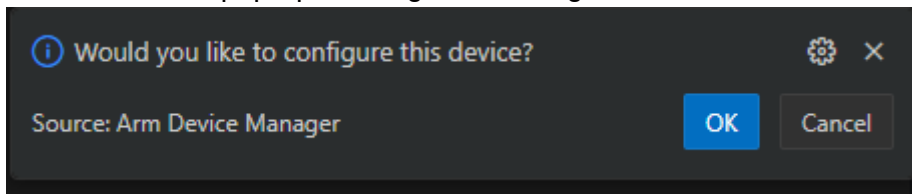
4. Click Device Manager in the Activity Bar to open the Device Manager.



- Connect device to your computer over USB.
The Device Manager detects the board and displays a pop-up message.
Press OK in the pop-up message and use it.



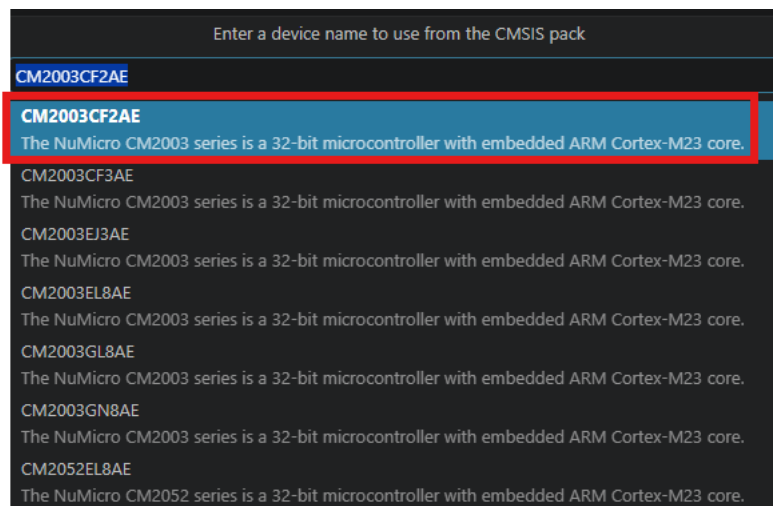
- Press OK in the pop-up message and configure this device.



- Text “nuvoton” in search bar and select CMSIS pack for the device.

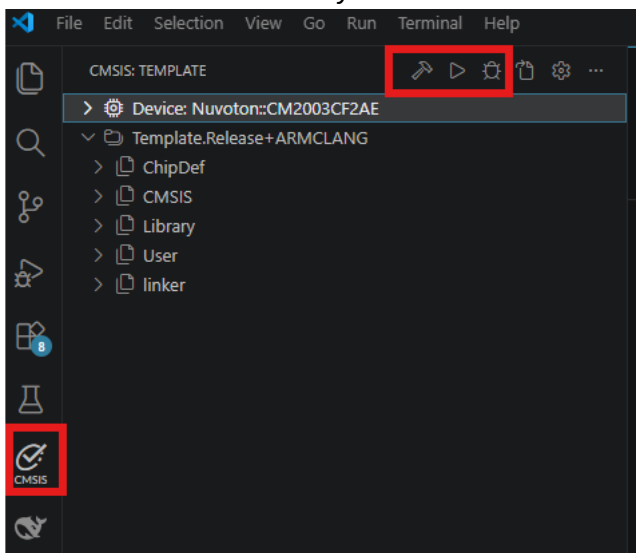


- Text “device model number” in search bar.

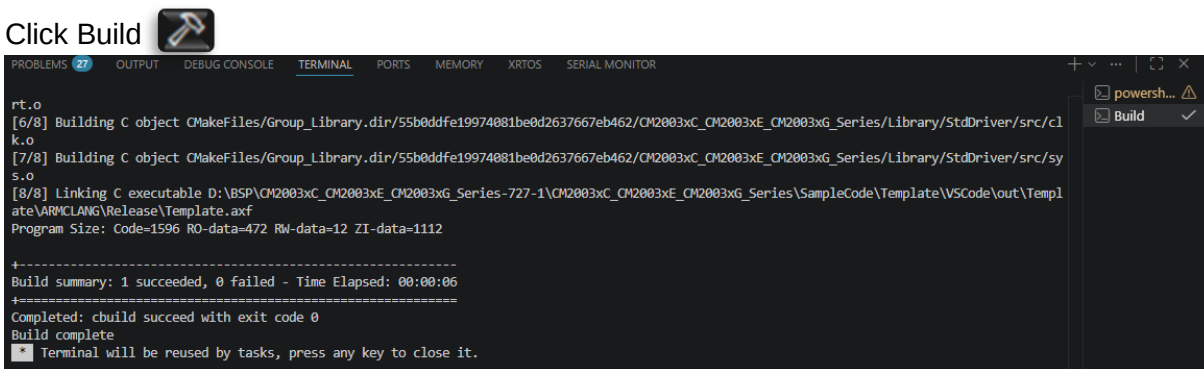


5 RUN THE EXAMPLE PROJECT

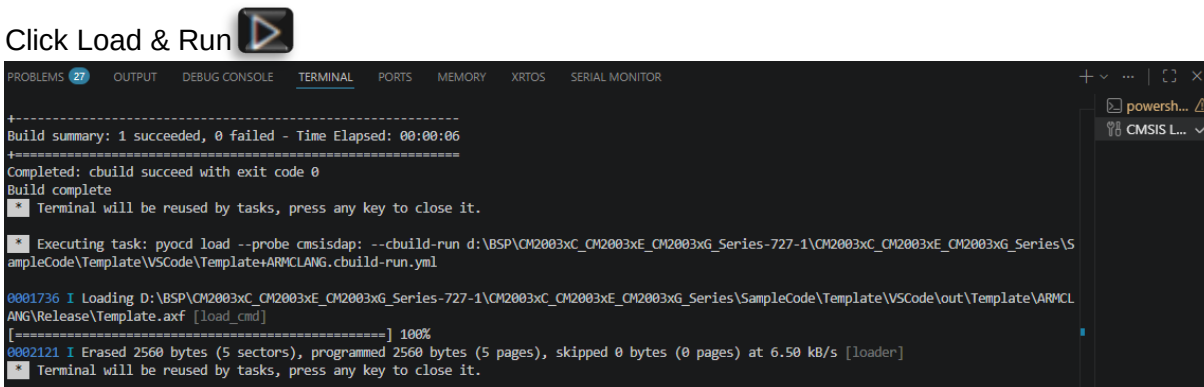
1. Click CMSIS in the Activity Bar



2. Click Build



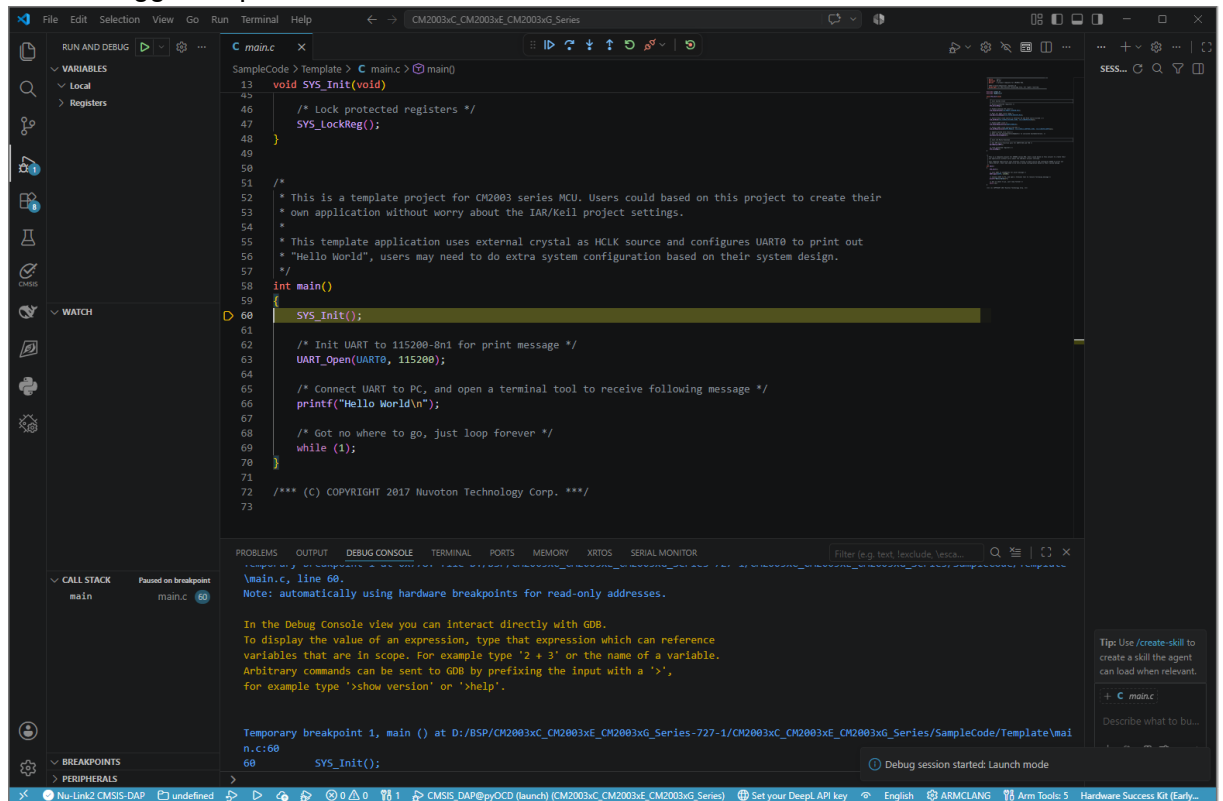
3. Click Load & Run



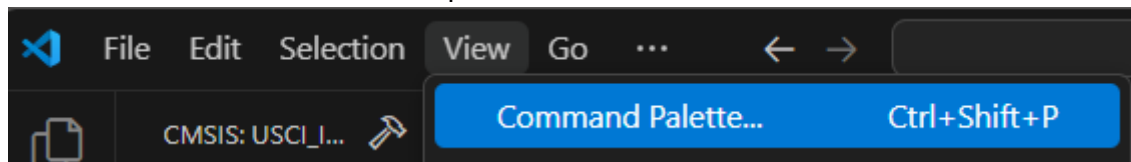
4. Click Load & Debug



The debugger stops at the main function.

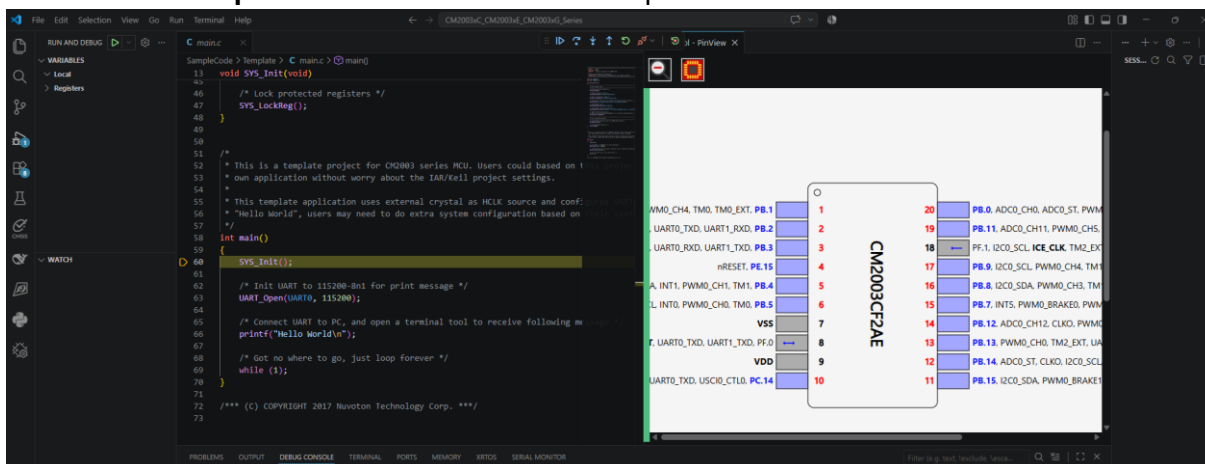


- Click View and select "Command palette" in the toolbar.



- Text "Nuvoton:Open PinView" in search bar to open PinView tool.

- Text “Nuvoton:Open PinLCD” in search bar to open PinLCD tool.



REVISION HISTORY

Date	Revision	Description
2026.08.04	3.01	1. Initially version.

Important Notice

Nuvoton Products are neither intended nor warranted for usage in systems or equipment, any malfunction or failure of which may cause loss of human life, bodily injury or severe property damage. Such applications are deemed, "Insecure Usage".

Insecure usage includes, but is not limited to: equipment for surgical implementation, atomic energy control instruments, airplane or spaceship instruments, the control or operation of dynamic, brake or safety systems designed for vehicular use, traffic signal instruments, all types of safety devices, and other applications intended to support or sustain life.

All Insecure Usage shall be made at customer's risk, and in the event that third parties lay claims to Nuvoton as a result of customer's Insecure Usage, customer shall indemnify the damages and liabilities thus incurred by Nuvoton.

*Please note that all data and specifications are subject to change without notice.
All the trademarks of products and companies mentioned in this datasheet belong to their respective owners.*